7th Grade Math · Weekly Check · Unit 4 C-GR.1.2

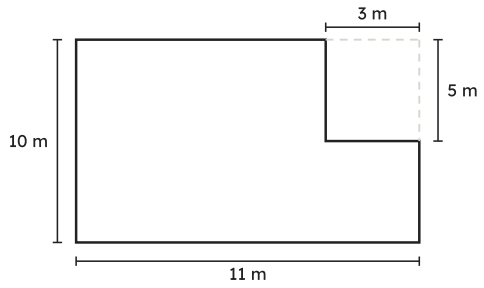
Benchmark: **MA.7.GR.1.2** · Area of polygons and composite figures Student code: _____ Calculator: yes

Learning goal: I can find the area of polygons and composite figures. (circle your level after scoring)

4	I can solve a real-world problem with a composite figure, including one with a region removed.
3	I can find the area of a polygon or composite figure by adding or subtracting areas.
2	I can split a figure into shapes I know.
1	I need help to start this benchmark.

1. A garden is shaped like the figure shown: an 11 m by 10 m rectangle with a 3 m by 5 m rectangular corner removed. The dashed lines show the part that was removed.

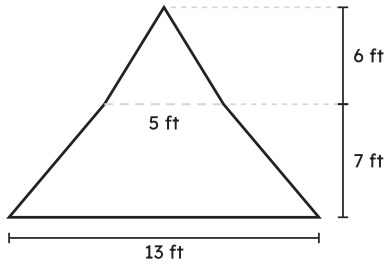
What is the area of the garden, in square meters?



Not drawn to scale

2. The front of a tent is shaped like the figure shown: a trapezoid with bases of 13 ft and 5 ft and a vertical height of 7 ft, with a triangle on top. The triangle's base is the 5 ft top base of the trapezoid, and its vertical height is 6 ft.

What is the area of the front of the tent, in square feet?



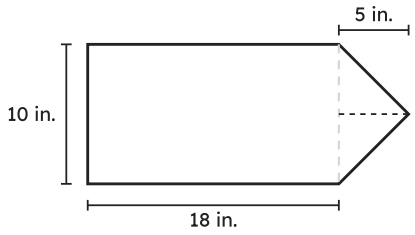
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Turn over →

Benchmark: **MA.7.GR.1.2** · Area of polygons and composite figures Student code: _____

3. A tabletop is shaped like the figure shown: an $18\,\text{in.} \times 10\,\text{in.}$ rectangle with a triangle attached to one end. The triangle's base is the 10 in. side and its height is 5 in.

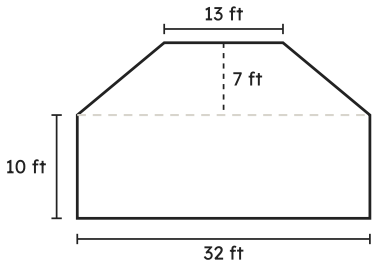
What is the area of the tabletop, in square inches?



Not drawn to scale

4. The end wall of a barn is shaped like the figure shown: a rectangle 32 ft wide and 10 ft tall, with a trapezoid on top. The trapezoid's bases are the 32 ft top edge of the rectangle and a 13 ft top side, and its vertical height is 7 ft.

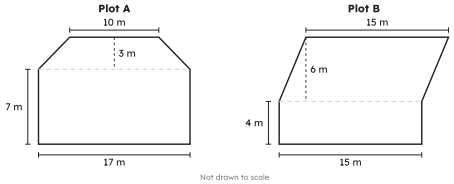
What is the area of the end wall, in square feet?



Not drawn to scale

5. Two garden plots are shown. Plot A is a trapezoid with bases of 17 m and 10 m and a vertical height of 3 m, joined along its longer base to a rectangle 17 m by 7 m. Plot B is a parallelogram with a base of 15 m and a vertical height of 6 m, joined along its base to a rectangle 15 m by 4 m.

Which plot has the greater area, and how much greater is it?



Not drawn to scale

Score: ____ / 5 My level: 1 2 3 4 → copy both to your progress strip

Answer key — C-GR.1.2 · MA.7.GR.1.2 · Area of polygons and composite figures

Unit 4: Area. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular $95\,\text{m}^2$

Whole rectangle: $11 \times 10 = 110\,\text{m}^2$.
Removed corner: $3 \times 5 = 15\,\text{m}^2$.
Area: $110 - 15 = 95\,\text{m}^2$.

2. Regular $78\,\text{ft}^2$

Trapezoid: $\frac{1}{2}(13 + 5)(7) = 63\,\text{ft}^2$.
Triangle: $5 \times 6 \div 2 = 15\,\text{ft}^2$.
Total: $63 + 15 = 78\,\text{ft}^2$.

3. Regular $205\,\text{in.}^2$

Rectangle: $18 \times 10 = 180$.
Triangle: $\frac{1}{2} \times 10 \times 5 = 25$.
Total: $180 + 25 = 205\,\text{in.}^2$.

4. Regular $477.5\,\text{ft}^2$

Rectangle: $32 \times 10 = 320\,\text{ft}^2$.
Trapezoid: $\frac{1}{2}(32 + 13)(7) = 157.5\,\text{ft}^2$.
Total: $320 + 157.5 = 477.5\,\text{ft}^2$.

5. Rigor **Plot A, by $9.5\,\text{m}^2$. Score: 1 point — the plot and the difference; the plot alone earns 0.**

Plot A: trapezoid $\frac{1}{2}(17 + 10)(3) = 40.5\,\text{m}^2$, rectangle $17 \times 7 = 119\,\text{m}^2$, so $159.5\,\text{m}^2$.
Plot B: parallelogram $15 \times 6 = 90\,\text{m}^2$, rectangle $15 \times 4 = 60\,\text{m}^2$, so $150\,\text{m}^2$.
$159.5 - 150 = 9.5$, so Plot A is greater by $9.5\,\text{m}^2$.

Re-check strip (3 items, C-GR.1.2-R)

1. Regular $108\,\text{ft}^2$

Whole rectangle: $12 \times 11 = 132\,\text{ft}^2$.
Removed notch: $6 \times 4 = 24\,\text{ft}^2$.
Area: $132 - 24 = 108\,\text{ft}^2$.

2. Regular $70\,\text{in.}^2$

Trapezoid: $\frac{1}{2}(10 + 8)(6) = 54\,\text{in.}^2$.
Triangle: $8 \times 4 \div 2 = 16\,\text{in.}^2$.
Total: $54 + 16 = 70\,\text{in.}^2$.

3. Regular $117\,\text{ft}^2$

Rectangle: $18 \times 6 = 108$.
Triangle: $\frac{1}{2} \times 6 \times 3 = 9$.
Total: $108 + 9 = 117\,\text{ft}^2$.

Comparison strip (2 items, C-GR.1.2-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular $165\,\text{ft}^2$

Whole rectangle: $15 \times 13 = 195\,\text{ft}^2$.
Removed corner: $5 \times 6 = 30\,\text{ft}^2$.
Area: $195 - 30 = 165\,\text{ft}^2$.

2. Regular $850\,\text{in.}^2$

One trapezoid: $\frac{1}{2}(32 + 18)(17) = 425\,\text{in.}^2$.
The two trapezoids are congruent, so the hexagon is $2 \times 425 = 850\,\text{in.}^2$.

Reteach plan — C-GR.1.2

MA.7.GR.1.2 · Area of polygons and composite figures

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

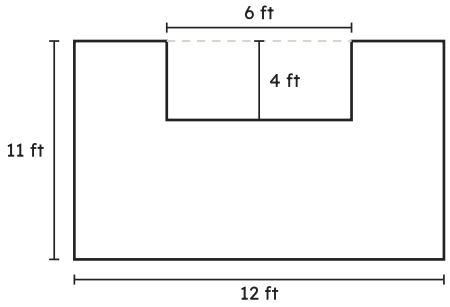
Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-GR.1.2-R

Benchmark: **MA.7.GR.1.2** · Area of polygons and composite figures Student code: _____ Calculator: yes

1. A kitchen floor is shaped like the figure shown: a 12 ft by 11 ft rectangle with a 6 ft by 4 ft rectangular notch removed from the middle of one edge. The dashed lines show the part that was removed.

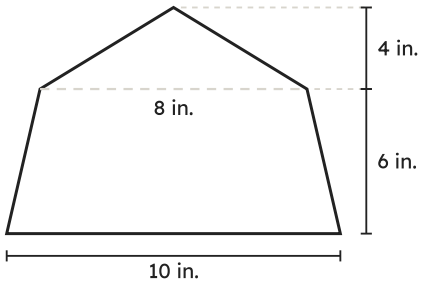
What is the area of the floor, in square feet?



Not drawn to scale

2. A stained-glass panel is shaped like the figure shown: a trapezoid with bases of 10 in. and 8 in. and a vertical height of 6 in., with a triangle on top. The triangle's base is the 8 in. top base of the trapezoid, and its vertical height is 4 in.

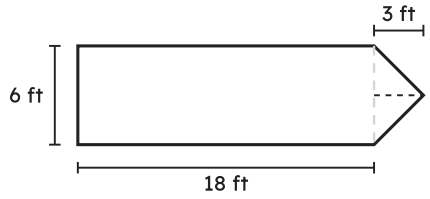
What is the area of the panel, in square inches?



Not drawn to scale

3. A garden bed is shaped like the figure shown: an $18\,\text{ft} \times 6\,\text{ft}$ rectangle with a triangle attached to one end. The triangle's base is the 6 ft side and its height is 3 ft.

What is the area of the garden bed, in square feet?



Not drawn to scale

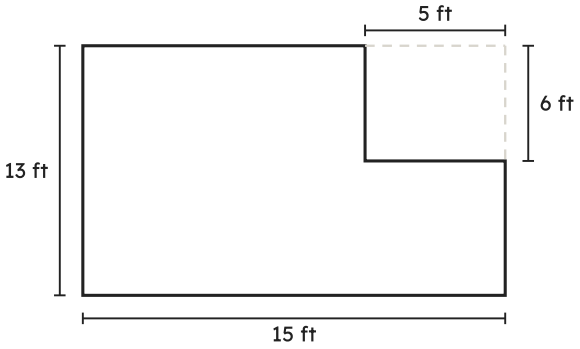
Score: ____ / 3 → copy to your progress strip

7th Grade Math · Comparison items (2) C-GR.1.2-X

Benchmark: **MA.7.GR.1.2** · Area of polygons and composite figures Student code: _____ Calculator: yes

1. A kitchen floor is shaped like the figure shown: a 15 ft by 13 ft rectangle with a 5 ft by 6 ft rectangular corner removed. The dashed lines show the part that was removed.

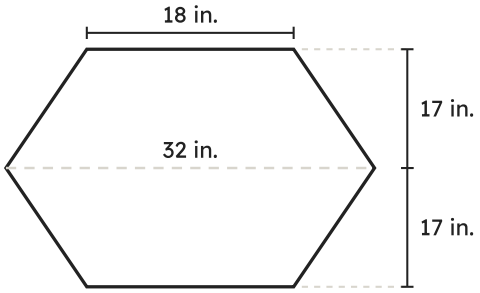
What is the area of the floor, in square feet?



Not drawn to scale

2. A hexagonal tabletop is shaped like the hexagon shown. It is made of two congruent trapezoids that share the 32 in. base drawn as a dashed line. Each trapezoid has a vertical height of 17 in., and its other base is 18 in.

What is the area of the tabletop, in square inches?



Not drawn to scale

Score: ____ / 2 → copy to your progress strip